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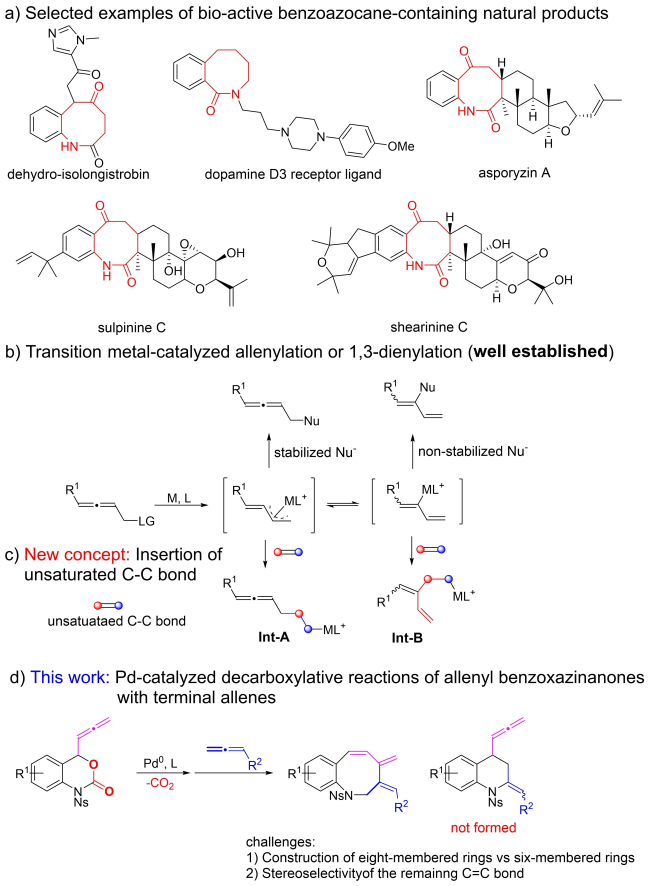
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Palladium-Catalyzed [6+2] Double Allene Annulation for Benzazocines Synthesis

*Haibo Xu and Shengming Ma**

Abstract: An efficient double allene protocol for the formation of benzazocines has been developed. The reaction constitutes a highly regioselective palladium-catalyzed formal [6+2] annulation of allenyl benzoxazinanones with terminal allenes forming the challenging 8-membered cycles. Decent yields and excellent regioselectivity have been observed under mild conditions with a remarkable *Z*-stereoselectivity for the exo-cyclic C=C bonds. The synthetic potentials of benzazocine products have been demonstrated.

Eight-membered cycles are attractive structural motifs in natural products and small molecule drugs.[1] Taking benzazocine as an example, it serves as the core structure of a variety of bio-active natural products, such as dehydro-isolongistrobin,[1a] sulpinine C,[1b] shearinine C,[1c] and asporyzin A[1i] (Scheme 1a). The limited flexibility increases the binding affinity of eight-membered cycles, while their well-defined conformations improve the physicochemical properties.[2] It is well-known that unfavorable enthalpic and entropic factors have been the major stumbling block for the synthesis of the eight-membered rings, thus much attention has been focused on this topic.[3,4] On the other hand, allenes bearing a leaving group at α-position could react with transition metal catalysts to generate the methylene-π-allylic metal intermediate,[5] which may readily undergo substitution reaction with stabilized nucleophiles or non-stabilized nucleophiles to give either allenes[6,7] and/or conjugated 1,3-dienes,[8] respectively (Scheme 1b). With our longstanding interest in developing new catalytic reactions of 2,3-allenyl derivatives,[6c–f,7e–l,8m] we envisaged that an insertion reaction of such methylene-π-allylic metal intermediates with an unsaturated C–C bond would generate intermediates **Int-A** and/or **Int-B** (Scheme 1c). Herein, we wish to demonstrate the first example of such a concept: the unexpected Pd-catalyzed formal [6+2] annulation of allenyl benzoxazinanones with allenes providing benzazocine derivatives highly efficiently with an exclusively *Z* stereoselectivity for non-terminal exo-cyclic C=C bond via **Int-B** (Scheme 1d).[9,10] The most favored 6-membered products via **Int-A** has not been formed.



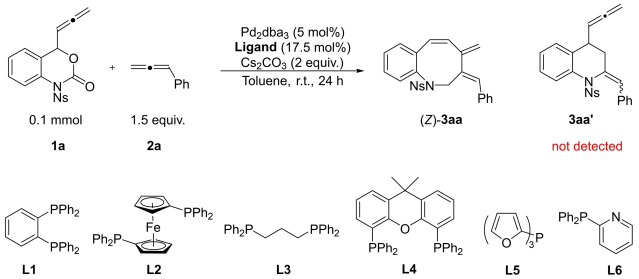
Scheme 1. Selected examples of naturally occurring benzazocines and transition metal-catalyzed reactions of 2,3-allenol derivatives.

Initially, we employed the allenyl benzoxazinanone **1a** and 1-phenylpropadiene **2a** as the model substrates. Unfortunately, when bidentate phosphorous ligands **L1-L4** were used, no product was detected (Table 1, entries 1–4). Interestingly, the use of monophosphine ligand, PPh_3, produced the non-favorable eight-membered ring product (*Z*)-**3aa** in 30 % yield, and interestingly the formation of the expected six-membered product **3aa′** was not observed (Table 1, entry 5). Further screening of mono-phosphine ligands showed that tri(fur-2-yl)phosphine **L5** delivered a

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Table 1: Optimization of the reaction conditions.



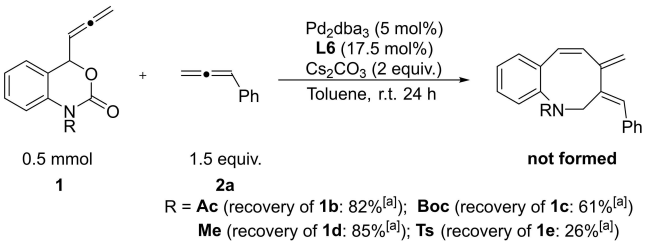
Entry	Ligand	Base	Yield of (*Z*)-**3aa** [%][a]	Recovery of **1a** [%][a]
1[b]	**L1**	Cs_2CO_3	0	25
2[b]	**L2**	Cs_2CO_3	0	26
3[b]	**L3**	Cs_2CO_3	0	24
4[b]	**L4**	Cs_2CO_3	0	0
5[b]	PPh_3	Cs_2CO_3	30	0
6[b]	**L5**	Cs_2CO_3	50	0
7[b]	**L6**	Cs_2CO_3	70	0
8	**L6**	Cs_2CO_3	82	0
9[c]	**L6**	Cs_2CO_3	48	0
10	**L6**	K_3PO_4	61	0
11	**L6**	NaOAc	27	0
12	**L6**	Na_2CO_3	63	0
13	**L6**	K_2CO_3	71	0
14	**L6**	Li_2CO_3	74	0
15[d]	**L6**	Cs_2CO_3	86(80[e])	0

[a] Determined by ^{1}H NMR analysis with CH_3NO_2 as the internal standard. [b] 10 mol% of Pd catalyst and 20 mol% of ligand were used. [c] 2.5 mol% of Pd catalyst and 8.75 mol% of ligand were used. [d] The reaction was conducted on 0.5 mmol scale. [e] Isolated yield.

better result and diphenyl-2-pyridylphosphine **L6** is the optimal ligand, providing the corresponding eight-membered product (*Z*)-**3aa** in 70 % yield (Table 1, entries 6 and 7). Reducing the catalyst loading and adjusting the ratio of catalyst and ligand increased the yield of (*Z*)-**3aa** to 82 % (Table 1, entry 8). Further reducing of the catalyst loading caused a dramatic drop of yield (Table 1, entry 9). Then we tried different base such as K_3PO_4, NaOAc, Na_2CO_3, K_2CO_3 and Li_2CO_3, and Cs_2CO_3 was still the optimal choice (Table 1, entries 10–14). Finally, the yield of (*Z*)-**3aa** was further improved to 86 % when the reaction was carried out on a 0.5 mmol scale (Table 1, entry 15).

Subsequently, the effect of the *N*-protecting group was investigated: When Ac (**1b**), Boc (**1c**), Me (**1d**) and Ts (**1e**) protected allenyl benzoxazinanones were used, no desired product was observed with recovery of the corresponding allenyl benzoxazinanones (Scheme 2), indicating the importance of the Ns group. The electron-withdrawing character of Ns group may facilitate the oxidative addition of palladium with the C–O bond and stabilize the negative charge on nitrogen after the subsequent decarboxylation.

With the optimal reaction conditions in hand, diversified terminal allenes[11] were investigated to examine the scope of this reaction for benzazocine derivatives synthesis (Scheme 3). Terminal aryl allenes bearing electron-donating groups, such as methoxy (**2b**) and methyl (**2c**), electron-withdrawing as well as synthetically versatile groups, such as halogen (**2d**–**2g**), acetyl (**2h**), ester (**2i**) and cyano (**2j**), on



Scheme 2. Effect of the *N*-protecting group. [a] Determined by ^{1}H NMR analysis with CH_3NO_2 as the internal standard.

the aryl ring all proved to work for this reaction, providing the desired benzazocine derivatives (*Z*)-**3ab**-(*Z*)-**3aj** in 40–80 % yields. The phenyl ring could also be replaced with a β-(**2k**) or α-naphthyl (**2l**) group, affording the products (*Z*)-**3ak** and (*Z*)-**3al** in 50 % and 58 % yields, respectively. For those results with yields less than 50 %, it may be due to the lower reactivities of the allenes **2** causing unmatched reactivity leading to partial decomposition of allenyl benzoxazinones **1** during the reactions, causing the lower yields of reactions.

Furthermore, non-aryl terminal allenes also participated in the reaction smoothly: Functionalized allenes such as 2,3-allenoate (**2m**) and allene with a functionalized alkyl group (**2n**) also worked well to deliver the corresponding products (*Z*)-**3am** and (*Z*)-**3an** with 64 % and 68 % yields. Such a reaction has also been applied to the late-stage modification of eucalyptene A (**2o**),[12] affording the corresponding benzazocine derivative (*Z*)-**3ao** in 40 % yield.

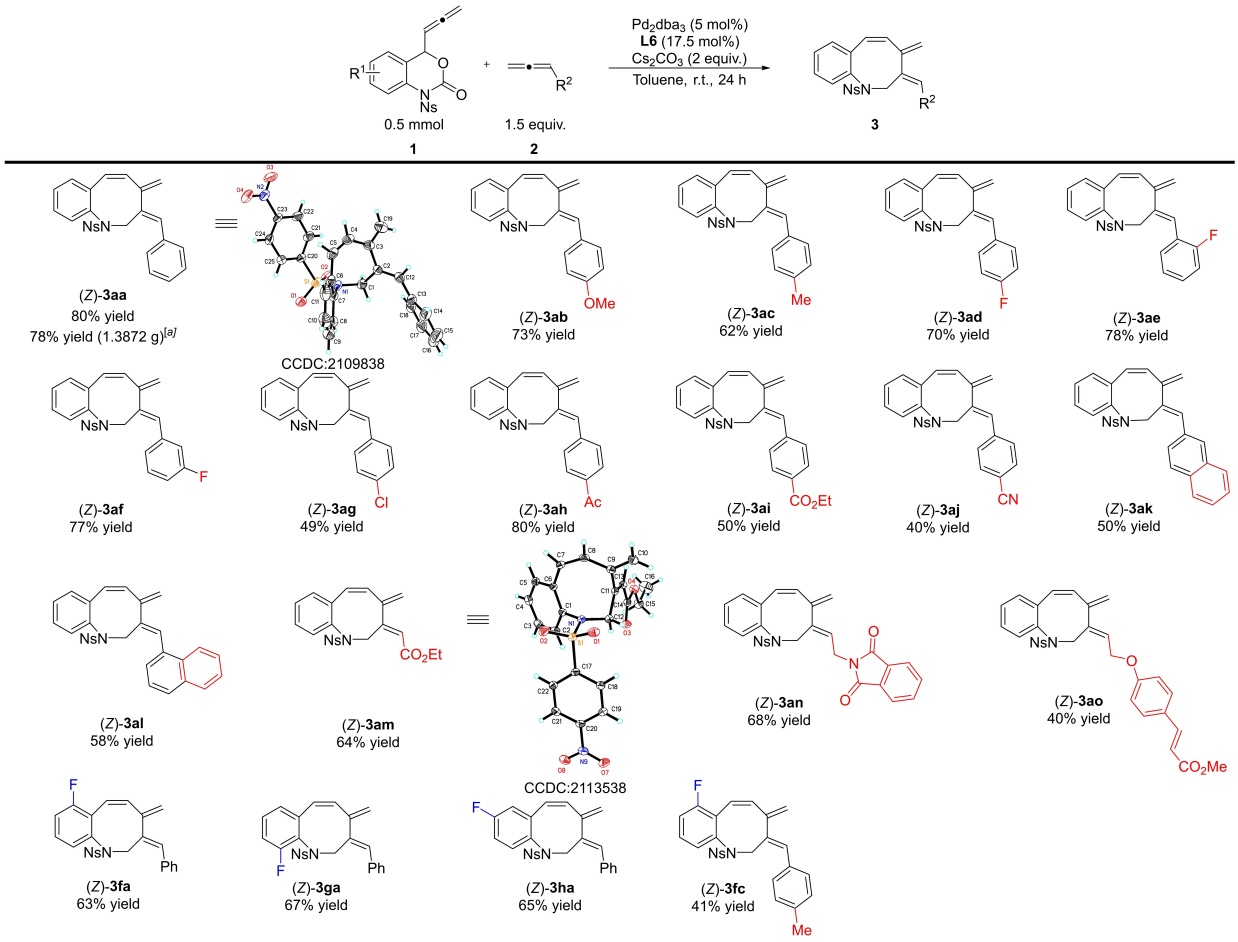
Considering the importance of fluorine atom in organic skeleton,[13] we turned to explore the scope of different fluorine-substituted allenyl benzoxazinanones **1f**–**1h**, yielding the desired benzazocines (*Z*)-**3fa**-(*Z*)-**3ha**, and (*Z*)-**3fc** in 41–67 % yields. The practicality of this method has been demonstrated by performing the reaction of allenyl benzoxazinanone **1a** and 1-phenylpropadiene **2a** on 4 mmol scale, providing benzazocine (*Z*)-**3aa** in 78 % yield (1.3872 g). The structures of (*Z*)-**3aa** and (*Z*)-**3am** were confirmed by X-ray single crystal analysis.[14]

The synthetic potential of this methodology has been demonstrated as shown in Scheme 4: the intermolecular Diels–Alder reaction of (*Z*)-**3ag** with PTAD (4-phenyl-1,2,4-triazoline-3,5-dione) at room temperature provided the tetracyclic benzazocine derivative **4** in 59 % yield;[15] interestingly, by conducting the reaction at 120 °C, (*Z*)-**3ag** could undergo a twofold Diels–Alder reaction with 3.0 equiv. of PTAD to afford hexacyclic product **5** with an excellent diastereoselectvity in 51 % yield. The structure of **5** was confirmed by X-ray single crystal analysis.[14]

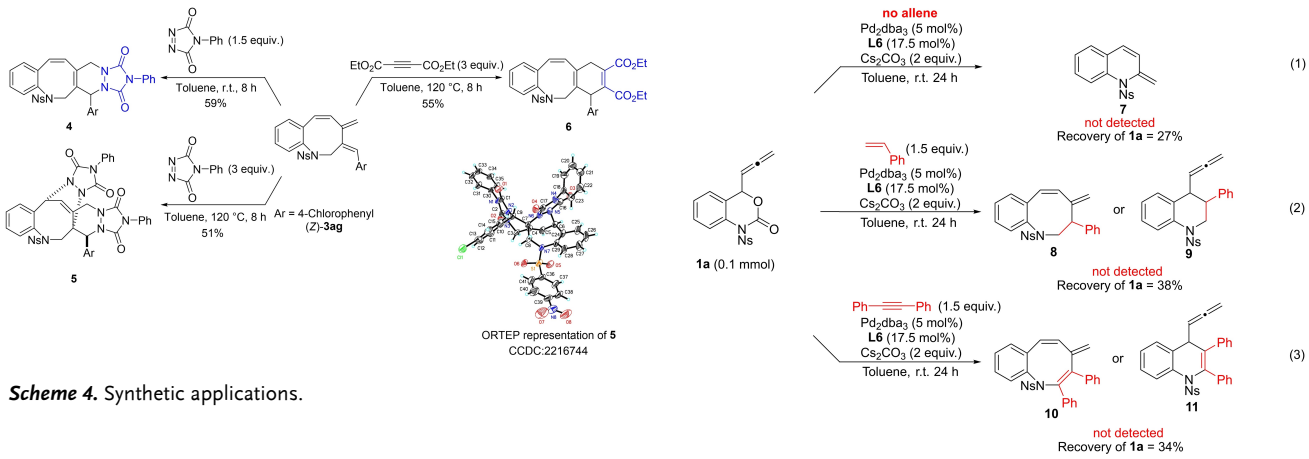
To gain insight into the mechanism, some control experiments have been carried out: When the reaction was conducted in the absence of terminal allene under the optimal conditions, the formation of the imagined six-membered product **7** derived from the reductive elimination of intermediate **III** (Scheme 6) was not observed (Scheme 5, eq 1), only recovery of **1a** was detected. Compared with the reaction described in entry 8 of Table 1, within the same

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Scheme 3. Scope of substrates. [a] The reaction was conducted using **1a** (4 mmol) and **2a** (1.5 equiv) in 20 mL of toluene.



Scheme 4. Synthetic applications.

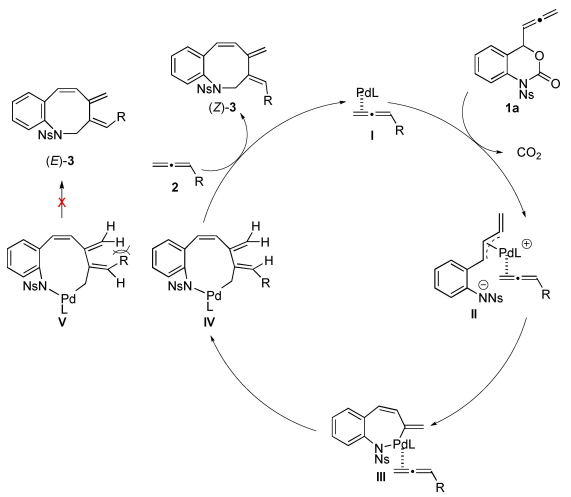
Scheme 5. Control experiments.

reaction time, the reaction rate of **1a** increased obviously after the addition of allene **2a**. It showed that allene **2a** did play an important role in promoting the decarboxylative oxidative addition of **1a**. Replacing the terminal allene with styrene or diphenylacetylene also led to recovery of **1a** only: neither eight-membered product **8**/**10** nor six-membered **9**/**11** was formed (Scheme 5, eqs 2 and 3). These results revealed the unique reactivity of terminal allene for this transformation.

A plausible mechanism is then proposed as shown in Scheme 6. Pd^0/L(allene **2**) species **I** may be the catalytically active species. Initially, it catalyzes the decarboxylation of **1a** to generate the methylene-π-allylpalladium zwitterion **II**,[5] which may isomerize to form more stable benzocyclic intermediate **III**. The highly regio- and stereoselective

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Scheme 6. Plausible mechanism for the highly selective formation of (*Z*)-**3**.

insertion of the terminal C=C bond in terminal allene **2** generating intermediate **IV** over **V** may be rationalized by the steric difference of the two C=C bond in allene and the steric hindrance between R and the olefinic H in intermediate **V**. Subsequent reductive elimination of intermediate **IV** in the presence of allene **2** would furnish the final product benzazocine (*Z*)-**3** and regenerates Pd/L(allene **2**) species **I** to complete this catalytic cycle. According to the data shown in Scheme 5, the pre-coordination of allene to the Pd atom is critical for this transformation. The favorable formation of the 8-membered product **3** over the 6-membered **3′** is caused by the exclusive formation of Intermediate **III**.

In conclusion, we have developed a completely new reactivity of methylene-π-allyl palladium species: the Pd-catalyzed [6+2] annulation of allenyl benzoxazinanones with the terminal C=C bond in allenes provides a new concept for the unexpected 8-membered ring formation affording functionalized benzazocine derivatives. The synthetic potentials of the products have been demonstrated. Control experiments suggested that neither alkene nor alkyne may replace allene to react with allenyl benzoxazinanone to deliver the related products. This new synthesis strategy will expedite the synthesis of eight-membered heterocycles for drug discovery and promote studies on new types of reactions of methylene-π-allyl palladium species. Currently we are actively working on the factors controlling the regioselectivity, the effect of ligand **L6**, and new reaction patterns.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Allene · Benzazocine · Palladium · [6+2] Annulation

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